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# Cheng Yuan (Ross) King

MSc AI | Data & ML Engineer

Portfolio: GitHub Portfolio
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## Career Aspiration

Seeking a full-time role in machine learning, data engineering, or AI infrastructure, contributing to scalable data pipelines, reliable ML workflows, cloud/data-centre-adjacent analytics, and production-quality model evaluation.

## Skills

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|---|---|
| **Programming** | Python, SQL, Bash, Java, TypeScript |
| **Data & ML** | PySpark, dbt, DuckDB, pandas, NumPy, scikit-learn, PyTorch, Hugging Face Transformers |
| **ML Methods** | model evaluation, feature engineering, cross-validation, hyperparameter tuning, classification, recommendation systems |
| **Infrastructure** | Linux, Slurm, HPC workflows, Git, Docker, GitHub Actions, CI, data testing |
| **Visualisation** | Streamlit, matplotlib, analytical dashboards, reporting marts |
| **Languages** | English, Mandarin Chinese (Native), Japanese (JLPT N1) |

## Selected Technical Projects

**Scalable Machine Learning with PySpark** **2026**
*PySpark, Python, Spark MLlib, Stanage HPC, Slurm* *University of Sheffield*

- Built distributed data mining and machine learning pipelines on datasets ranging from 1.9M to 20M records using Apache PySpark on the University of Sheffield Stanage HPC cluster.
- Implemented web log mining, traffic prediction, HIGGS classification, MovieLens recommendation, and clustering workflows using Spark SQL and Spark MLlib.
- Applied GLM, Logistic Regression, Random Forest, Gradient Boosted Trees, ALS, and k-means across classification, regression, recommendation, and segmentation tasks.
- Designed robust evaluation workflows using temporal splits, stratified sampling, cross-validation, hyperparameter tuning, and final held-out test comparison.

**UK Property Analytics Warehouse** **2026**
*SQL, dbt-core, DuckDB, Streamlit, GitHub Actions, Python* *Analytics Engineering Project*

- Built an end-to-end analytics warehouse over 4.99M UK Land Registry property transactions from 2021–2025, modelling raw data into staging, intermediate, fact, dimension, and reporting layers.
- Implemented dbt transformations with 88 data tests, source freshness checks, SQL linting, and GitHub Actions CI to enforce reliable data quality before merging.
- Developed reporting marts for regional year-on-year price change, postcode transaction volume, and new-build price premium analysis.
- Published generated dbt documentation and a Streamlit dashboard, with reproducible scripts for data download, DuckDB loading, and local warehouse rebuilds.

**LLM Event Extraction on GPU/HPC** **2026**
*Python, Qwen2.5-7B-Instruct, Hugging Face, NVIDIA A100, Slurm* *NLP/LLM Project*

- Developed a zero-shot LLM baseline for event extraction using Qwen2.5-7B-Instruct on MAVEN and WikiEvents datasets.
- Compared unconstrained and constrained-label prompting strategies for trigger detection, event type prediction, and argument extraction.
- Ran float16 inference on NVIDIA A100-SXM4-80GB GPU resources through the University of Sheffield Stanage HPC environment.
- Built evaluation and error-analysis scripts to measure valid JSON rate, trigger accuracy, type accuracy, combined accuracy, and prediction failure patterns.

## Education

**MSc Artificial Intelligence**, *University of Sheffield* Sep 2025 — Sep 2026
*Core modules: Scalable Machine Learning, Natural Language Processing, Parallel Computing with GPUs, Machine Learning, Data Science, Text Processing* Sheffield, UK

**BSc Computer Science**, *Queen's University Belfast* Sep 2021 — Jun 2024
*Core modules: Data Structures and Algorithms, Software Engineering, Advanced Computer Architecture, Cloud Computing* Belfast, UK

## Additional Information

- Additional portfolio projects include a shipped privacy-first browser file converter, a speech speed/tempo classification pipeline, a document question-answering assistant, and an LLM red-team evaluation harness in progress.
- Career interests include applied ML systems, data engineering, scalable computing, AI infrastructure, cloud/data-centre analytics, and reliable software engineering.